LAB 4 Conversation Export

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Section: 5H

This PDF contains the user prompts and assistant replies from the Breakout LAB-4 coding conversation, organized chronologically.

Conversation

User Prompt

Clone this repo into this folder: https://github.com/Shishir-Hegde14/LAB_4

Assistant Reply

Cloned https://github.com/Shishir-Hegde14/LAB_4 directly into C:\Users\shish\OneDrive\Desktop\LAB-4. The repository is on main tracking origin/main and contains .git, breakout, .gitignore, and README.md.

User Prompt

Task 1: In game folder, change the code such that when a ball touches a brick, the brick despawns and the ball rebounds as usual. Do not change the bounce-back mechanism; only make the brick despawn. Add a simple particle effect of breaking.

Assistant Reply

Updated game_engine.py so hit bricks despawn while keeping handle_ball_brick_collision unchanged. Added simple brick-colored break particles and renderer support for drawing them. Verified with compileall and a direct collision check: brick count dropped from 32 to 31, ball vy flipped from -5 to 5, and 8 particles spawned.

User Prompt

Task 2: In games folder, when brick scores become 0 show YOU WIN and end the game, with two white buttons: restart game and quit. Implement lives: give 3 lives per round, display lives on screen, and every time the ball falls below, remove a life and reset player/ball position while brick states persist. Once lives become 0, show GAME OVER and the same two buttons. Do not change the logic of any other module.

Assistant Reply

Added YOU WIN and GAME OVER states, two white end-screen buttons, 3 lives per round, life display, restart behavior, and quit behavior. Ball and paddle reset on life loss while bricks persist. Restart starts a fresh round. Verified compileall and direct checks for win, life loss, game over, and restart button behavior.

User Prompt

Task 3: Implement different types of bricks. Red: 1 hit with red sparkle. Moss: reddish with green texture, 3 hits, green sparkle on hit, green-red sparkle on break. Concrete: grey concrete effect, 5 hits, cracks more after each hit, grey sparkle on hit, grey-brown sparkle on break. Metal: shiny glowing metal, shake on hit, 6 hits, silver sparkle on hit, silver spike particles on break. Obsidian: purple glow, unbreakable, max 3, purple sparkle on hit. Randomize bricks each game with counts normal > moss > concrete > metal and max 3 obsidian. Do not change any other game mechanism.

Assistant Reply

Implemented brick types in brick.py, random spawn assignment in game_engine.py, type-specific hit/break behavior, particles, metal screen shake, and win counting that ignores unbreakable obsidian. Renderer now draws red, moss, concrete, metal, and obsidian visuals including cracks, glow, shine, and spike particles. Verified compileall, spawn counts, obsidian cap, hit counts, obsidian unbreakability, win with obsidian remaining, and render smoke tests.

User Prompt

Make the moss block fully green with dark green dots, make it look like a mossy block. Make the concrete block a bit of different shape if possible, slightly like an actual concrete block. Keep all mechanisms and sparkle effects the same.

Assistant Reply

Updated only renderer.py visuals: moss is now fully green with dark green dots/speckles, and concrete now has a slightly irregular concrete-block shape with bevel/shadow details. Mechanics and sparkle effects are unchanged. Verified compileall and render smoke test.

User Prompt

Task 4: Remove the bricks left counter. Add a score mechanism: normal block +1, moss +3, concrete +5, metal +6. Show this on screen. Improve UX of score and lives text. Show user score when game is over or user wins. Add combo mechanism: after breaking 3 blocks in a row, next 2 block breaks get +3 and show yellow 1X text particle. If combo continues to 5 blocks in a row, next 3 breaks get +4 and show red 2X text. For higher streaks show violet 5X and add +5 per break. Combo degradation: if ball bounces 5 times off a brick and does not break anything, reset combo. Change ball and paddle colors/effects for combo tiers. Shake screen on 1X to 2X or 2X to 5X transitions. Reduce particles slowly on non-combo hits while in combo. Lose combo on life loss. If ball is in 5X combo when it breaks the last block, make the ball explode and break all blocks including non-breakable blocks, then show restart/YOU WIN window.

Assistant Reply

Implemented score display, final score on win/game-over, combo scoring tiers, combo text particles, combo ball trails, paddle glow/color changes, transition screen shake, combo reset on life loss, combo degradation, and the 5X final-block explosion that removes all remaining blocks including obsidian before showing YOU WIN. Verified compileall, score/combo tests, combo decay, life-loss reset, final explosion/win delay, and render smoke test.

User Prompt

Make changes: the combo degradation should not happen if the ball is bouncing off other bricks. Only the bounce of the ball with the user bar should cause degradation.

Assistant Reply

Moved combo degradation so it only happens when the ball bounces off the paddle/user bar. Removed degradation from non-breaking brick and obsidian hits. Verified compileall, that brick hits no longer reduce combo, and that paddle bounce degradation still resets combo after 5 hits.

User Prompt

Export this conversation as a PDF, with all the prompts and the replies, properly. Add my name: shishir, my SRN: PES1UG24CS438, section: 5H in the PDF.

Assistant Reply

Generated this PDF export with the requested student details and the conversation prompts and replies in order.